

PREDICTION IS NOT THE PRODUCT.

RTO Shield converts COD order risk prediction into **financially optimal intervention decisions** for Indian e-commerce merchants — a measured, frozen, independently reproduced decision engine.

THE PROBLEM

Cash-on-delivery dominates Indian e-commerce, and **28.27% of test COD orders** came back as Return-to-Origin — each costing **₹150** in reverse logistics. Low-value risky orders are economically unsalvageable; high-value ones deserve a frictionless path.

THE SYSTEM

A calibrated LightGBM scores P(RTO) from 16 order-time features. An expected-loss engine prices every intervention per order —

$EL(action) = friction + P_{rto_after} \times ₹150 - P_{success} \times margin$ — and routes each order to the argmin: **ALLOW**, **VERIFY** (₹2 friction, –30% RTO), or **DEPOSIT** (–80% RTO).

THE PROOF

On a strictly held-out test window (7,174 COD orders), the multi-action policy earned **₹71,741 expected savings and ₹69,786 realized** against actual labels — –2.7% vs forecast, inside Monte Carlo noise. The uncalibrated router's forecast collapsed **–15.3%** against the same labels: calibration bought the only forecast that held up.

13.1%

COD PORTFOLIO
PROFIT UPLIFT

2.0x

SAVINGS VS BEST
SINGLE THRESHOLD

94.7%

OF MEASURED
BAYES CEILING

100%

P(SAVINGS > 0)
5,000 DRAWS

SAVINGS VS ALWAYS-ALLOW – HELD-OUT TEST WINDOW

